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Abstract

SAFORA (Smart Ride) is an AI-powered ride-hailing platform designed to revolutionize passenger safety in Pakistan's transportation sector. Unlike existing reactive solutions that rely on manual SOS buttons, SAFORA implements a proactive safety paradigm through three core innovations: The Pink Pass (biometrically verified women-only mode), The Safety Sentinel (automated GPS monitoring), and Smart Driver Selection (AI-driven matching). This document outlines the software requirements for the initial phase of the SAFORA project, detailing the architectural, functional, and non-functional specifications necessary for development.

The system employs a hybrid microservice architecture, combining a MERN stack (MongoDB, Express, React, Node.js) for the core application with a Python/Flask AI service for machine learning tasks. Key AI models include Linear Regression for price prediction, CNN for liveness verification, and K-Means clustering for demand heatmaps. By integrating these advanced technologies, SAFORA aims to address critical gaps in current ride-hailing services, specifically focusing on the safety of female passengers and the equitable distribution of rides among drivers.

Furthermore, this SRS document provides a comprehensive breakdown of the system's use cases, including user registration, ride management, and emergency response protocols. It establishes firmly the constraints, assumptions, and dependencies that will guide the development process. The ultimate goal of this specification is to serve as a blueprint for a robust, scalable, and secure platform that restores trust in public transportation through technological innovation.

1. INTRODUCTION and Background

1.1 Product (Problem Statement)

Current ride-hailing services in Pakistan, such as Uber, Careem, InDrive, and Yango, suffer from three critical safety and operational deficiencies that severely impact user trust and reliability. First, these platforms rely primarily on reactive safety measures, such as manual SOS buttons, which are often ineffective in acute emergencies where a victim may be physically unable to access their phone. Second, gender verification mechanisms are porous; women-only modes rely on self-declared gender data, allowing unauthorized users to exploit the feature, thereby compromising the safety of female passengers who specifically seek secure transport options.

Third, existing driver allocation algorithms prioritize proximity above all else, leading to a "winner-takes-all" dynamic where top-rated drivers monopolize requests while others sit idle. This lack of fairness affects driver retention and service quality. Furthermore, the absence of transparent pricing models often leads to disputes between passengers and drivers. SAFORA addresses these systemic failures by introducing a proactive safety paradigm, rigorous biometric verification, and a fairness-aware distribution algorithm.

1.2 Background

The ride-hailing domain in Pakistan is a rapidly expanding market, driven by urbanization and the lack of reliable public transport infrastructure. However, this growth faces significant headwinds due to safety concerns. High-profile incidents of route deviation, harassment, and theft have eroded public confidence. While international players have established a foothold, their global standard operating procedures often fail to account for the specific security challenges of the local environment. SAFORA is conceived against this backdrop as a locally engineered solution that leverages advanced Artificial Intelligence to provide a layer of security that acts before an incident occurs, rather than merely reporting it afterwards.

1.3 Scope

The scope of the SAFORA project encompasses the end-to-end development of a comprehensive ride-hailing ecosystem. This includes:

1. A cross-platform mobile application for Passengers (Android/iOS) featuring real-time booking, tracking, and the "Pink Pass" verification interface.
2. A dedicated mobile application for Drivers that manages ride requests, navigation, and earnings.
3. A centralized Backend Server responsible for orchestration, user management, and secure data handling.
4. A specialized AI Microservice that hosts the machine learning models for liveness detection, price prediction, and route monitoring.
5. An Administrative Dashboard for real-time fleet supervision and incident response.

Key features within scope include the Pink Pass biometric verification system, the Safety Sentinel automated route monitoring engine, and the smart driver matching algorithm. Explicitly out of scope for this initial phase are multi-city operations (launch is limited to Lahore), integration with third-party digital wallets (cash-only initially), and carpooling features.

1.4 Objectives

1. Enhance Passenger Safety: Implementation of automated route deviation detection capable of identifying anomalies greater than 500 meters within 30 seconds, reducing emergency response latency significantly compared to manual reporting.

2. Ensure Gender Verification Integrity: Achieve greater than 98% accuracy in liveness detection to effectively prevent identity spoofing, ensuring that the "Pink Pass" mode remains exclusive to verified female users.

3. Optimize Driver Assignment: Deploy a weighted matching algorithm that reduces average passenger wait times to under 3 minutes while ensuring that 80% of available drivers receive at least one ride per hour during peak periods.

4. Provide Transparent Pricing: Utilize linear regression models to predict fares with a variance of less than PKR 30, providing users with a detailed cost breakdown before they confirm their booking.

5. Enable Proactive Safety Monitoring: Establish a real-time monitoring loop that automatically triggers alerts to administrators and emergency contacts without requiring user intervention, shifting the safety paradigm from reactive to proactive.

1.5 Challenges

1. Real-time GPS Accuracy: Ensuring reliable deviation detection is technically challenging due to GPS drift and signal loss in dense urban environments. The system must filter noise from actual anomalies to prevent false alarms.

2. AI Model Latency: Running complex computer vision models for liveness detection on mobile devices or transmitting video data to a server introduces latency. Optimizing these models for near-instantaneous response is critical for user experience.

3. Network Instability: Pakistan's mobile network infrastructure can be inconsistent. The application must be robust enough to handle intermittent connectivity without losing critical state data or failing to transmit safety alerts.

4. Privacy versus Security: Balancing the need for intrusive biometric verification and continuous location tracking with user privacy rights and data protection regulations requires careful architectural planning and encryption implementation.

1.6 Learning Outcomes

The development of SAFORA offers a diverse range of technical and professional learning outcomes. Technically, the team will gain deep expertise in microservices architecture, integrating heterogeneous systems (Node.js and Python), and deploying machine learning models in a production environment. We will master real-time geospatial data processing using technologies like Socket.io and Turf.js. Professionally, the project fosters skills in agile project management, requirements engineering, and the implementation of industry-standard security practices such as JWT authentication and AES-256 data encryption.

1.7 Nature of End Product

The final deliverable is a production-ready software suite consisting of two mobile applications (Passenger and Driver) and a web-based administrative console. These interfaces are powered by a robust backend infrastructure deployed on cloud servers. The end product will support real-world ride-hailing workflows, from account creation and identity verification to ride completion and payment. It is not merely a prototype but a functional system capable of handling concurrent users, processing real-time payments, and executing safety protocols in a live environment.

1.8 Completeness Criteria

The project will be deemed complete upon the satisfaction of the following rigorous criteria:

1. **Functional Integrity:** Use cases for all three user roles (Passenger, Driver, Admin) must be fully operational without critical bugs.
2. **Safety Validation:** The Safety Sentinel must successfully trigger alerts in 100% of tested simulated route deviations greater than the threshold.
3. **Biometric Accuracy:** The Pink Pass system must reject at least 95% of spoofing attempts (photos/videos) while accepting valid live users.
4. **Performance Benchmarks:** The system must support 100 concurrent uniform users with API response times remaining under 500ms.
5. **Documentation:** Comprehensive technical manuals, user guides, and API documentation must be delivered.

1.9 Business Goals

From a business perspective, SAFORA aims to disrupt the local market by capturing 15% of the female ride-hailing demographic in Lahore within the first year of operation. The platform targets a 40% reduction in safety-related support tickets compared to industry averages, translating to lower operational costs and higher brand loyalty. Furthermore, the verified safety features are positioned to establish B2B partnerships with corporate entities seeking safe travel options for their female workforce, creating a sustainable revenue stream beyond individual fares.

1.10 Literature Review

A comprehensive review of existing literature and competitive platforms reveals significant gaps. While Uber and Careem dominate the market, their safety features are largely reactive. Research by Zhang et al. (2021) in the "Journal of Transportation Safety" demonstrates that machine learning models can predict driver anomalies with 92% accuracy using GPS data, a capability currently absent in major local apps. Similarly, Patel et al. (2020) in "IEEE Transactions on Biometrics" confirm the efficacy of CNN-based liveness detection in preventing identity fraud, validating SAFORA's architectural choice for the Pink Pass. Comparison with "Her Way" in South Africa shows that while gender-segregated services exist, SAFORA's addition of automated biometric verification represents a novel technological advancement in this niche.

1.11 Document Conventions

This document adheres to strict typographical conventions to ensure clarity. Italicized terms denote external systems or third-party APIs (e.g., **Google Maps API, Twilio**).

Bold text is used to highlight key terms, identifiers (e.g., **UC-01**), and critical requirements. The document follows the IEEE 830-1998 standard for structure. Tense usage implies binding requirements ("shall") versus non-binding recommendations ("should"). All currency figures are cited in Pakistani Rupees (PKR) unless otherwise noted.

2. Overall Description

2.1 Product Features

SAFORA provides a comprehensive suite of features designed for safety and efficiency. **1. Pink Pass:** A specialized mode for female passengers that enforces strict biometric checks to ensure that only verified women can request or accept these rides.

2. Safety Sentinel: An always-on background service that monitors GPS coordinates against the planned route, automatically triggering alerts for unexplained deviations or stops.

3. Smart Driver Selection: A multifaceted matching engine that balances proximity with driver ratings and fairness metrics to optimize fleet utilization.

4. Dynamic Price Prediction: A transparency tool that provides users with a fair fare estimate based on distance, time, and current demand, reducing negotiation friction.

5. Admin Command Center: A dashboard providing real-time visibility into the entire network, including live ride tracking, heatmap visualizations of demand, and instant access to safety alerts.

2.2 User Classes and Characteristics

Passenger: The primary consumer of the service. Includes the general public, aged 18-65, who require reliable transport. They are assumed to have basic smartphone literacy. A critical subset is the **Pink Pass User**, female passengers who prioritize safety and are willing to undergo additional verification steps.

Driver: Service providers who own their vehicles. They are motivated by income stability and fair treatment. They possess moderate technical skills, capable of using navigation apps. Key requirements include valid licensure and vehicle registration.

Administrator: Internal staff responsible for platform health. They are technically proficient, capable of interpreting data analytics, and trained in emergency response protocols. They operate the dashboard to monitor safety compliance and manage user disputes.

2.3 Operating Environment

Client Layer: The mobile applications are built to run on Android devices (version 8.0 Oreo and above) and iOS devices (version 13.0 and above), ensuring broad compatibility with the local device market.

Server Layer: The backend infrastructure is hosted on cloud services (AWS/DigitalOcean/Heroku), utilizing Node.js for the API Gateway and Python for the AI Microservice.

Data Layer: MongoDB v6.0+ is used for flexible document storage, hosted on MongoDB Atlas for high availability.

Network: The system requires a continuous internet connection, functioning effectively on 4G/LTE networks and maintaining stability on 3G connections common in developing areas.

2.4 Design and Implementation Constraints

Regulatory: The system must strictly adhere to the draft Personal Data Protection Bill of Pakistan, particularly regarding the encryption of biometric data and CNIC information. SMS notifications must comply with PTA guidelines.

Technical: The solution is constrained by the need to use Google Maps API, which has associated costs; the architecture must optimize API calls to stay within budget. Mobile battery usage for GPS tracking must be optimized to not exceed 15% drain per hour.

Economic: As an FYP, the budget is limited. The system design prioritizes free-tier services and open-source technologies (MERN stack, Flask) to maintain financial viability during the development phase.

2.5 Assumptions and Dependencies

Assumptions: It is assumed that end-users possess smartphones with functional GPS sensors and cameras. We assume that the GPS accuracy provided by consumer devices is within a 10-meter margin of error. It is also assumed that the cellular network will provide at least 90% uptime in the target operational area.

Dependencies: The system is critically dependent on the Google Maps API for routing and geocoding services. It relies on Twilio for the delivery of SMS alerts; any outage in these third-party services will impact core functionality. The AI models depend on the availability of sufficient training data to achieve the targeted accuracy rates.

3. Functional Requirements

All functional requirements are detailed below as formal Use Cases.

3.1 Use Case Descriptions

The functionality of SAFORA is encapsulated in the following use cases, categorized by priority:

Critical Use Cases (High Priority)

UC-01: User Registration (Passenger) - Account creation and identity verification.

UC-02: User Registration (Driver) - Driver onboarding and document submission.

UC-03: User Authentication - Secure login via JWT.

UC-04: Request Ride - Booking flow with locational inputs.

UC-05: Accept Ride (Driver) - Ride acceptance logic.

UC-06: Track Ride in Real-time - GPS streaming and map visualization.

UC-07: Complete Ride - Fare calculation and payment.

UC-08: Pink Pass Enrollment - Biometric video submission.

UC-09: Pink Pass Liveness Verification - AI-based anti-spoofing check.

UC-10: Route Deviation Detection - Automated safety monitoring.

UC-11: Trigger Safety Alert - Emergency response activation.

Important Use Cases (Medium Priority)

UC-12: Driver Matching Algorithm - Intelligent dispatching.

UC-13: Price Prediction - Fare estimation.

UC-14: Rate Driver/Passenger - Reputation system.

UC-15: Submit Feedback - Qualitative data collection.

UC-16: Admin Monitor Dashboard - System oversight.

UC-17: Generate Demand Heatmap - Analytics visualization.

Supporting Use Cases (Lower Priority)

UC-18: View Ride History - User activity logs.

UC-19: Manage Payment Methods - Cash/Digital wallet settings.

UC-20: Update Profile - User data management.

3.2 Detailed Use Cases

UC-01: User Registration (Passenger)

Identifier	UC-01
Name	User Registration (Passenger)
Priority	High
Actors	Passenger, System, AI Service
Pre-conditions	User has downloaded the SAFORA mobile app
Post-conditions	User account created and ready for ride booking

Typical Course of Action:

Actor Action	**System Response**
1. User opens app and selects "Sign Up as Passenger"	2. System displays registration form
3. User enters input	4. System validates input format
5. User uploads CNIC	6. System performs OCR verification
7. User submits form	8. System sends OTP
9. User enters OTP	10. System verifies OTP
	11. System creates user account...
	12. System sends welcome email...
	13. System displays success message...

Alternative Courses:

- • A1: Invalid email format Display error
- • A2: Phone number already registered
- • A3: CNIC OCR mismatch
- • A4: OTP expired

UC-04: Request Ride

Identifier	UC-04
Name	Request Ride
Priority	High
Actors	Passenger, System, AI Service, Driver
Pre-conditions	Passenger is logged in; GPS enabled
Post-conditions	Ride request created and sent

Typical Course of Action:

Actor Action	**System Response**
1. Passenger opens "Book Ride"	2. System detects GPS location
3. Passenger sets pickup	4. System validates location
5. Passenger sets destination	6. System calculates route
	7. System sends to AI Service...
	8. AI Service predicts price...
	9. System displays price...
10. Passenger selects ride type	11. If Pink catch check verified...
12. Passenger confirms	13. System finds nearby drivers...
	14. System sends request...
	15. System displays "Finding driver"...
	16. Driver receives notification

Alternative Courses:

- • A1: No GPS signal
- • A2: Destination outside service area
- • A3: Pink Pass not verified
- • A4: No drivers available
- • A5: Driver rejects -> Next driver

UC-05: Accept Ride (Driver)

Identifier	UC-05
Name	Accept Ride (Driver)
Priority	High
Actors	Driver, System, Passenger
Pre-conditions	Driver is online; ride request received
Post-conditions	Ride is active

Typical Course of Action:

Actor Action	**System Response**
1. (Precondition met)	2. Driver app alert sound
3. Driver views request	4. System starts countdown
5. Driver taps "Accept"	6. System changes status to "accepted"
	7. System notifies passenger...
	8. System displays driver details...
	9. System provides navigation...
	10. System starts GPS tracking (UC-06)

Alternative Courses:

- • A1: Driver taps "Reject"
- • A2: Timer expires
- • A3: Network disconnection

UC-08: Pink Pass Enrollment

Identifier	UC-08
Name	Pink Pass Enrollment

Priority	High
Actors	Female Passenger, System, AI Service
Pre-conditions	User gender "Female"
Post-conditions	Pink Pass activated

Typical Course of Action:

Actor Action	**System Response**
1. Passenger navigates to Profile	2. System displays info
3. Taps "Enroll Now"	4. System displays consent
5. Accepts consent	6. System activates camera
	7. System instructions: "Look and blink"
8. Positions face and blinks	9. System captures video
	10. Sends to AI (UC-09)...
	11. AI performs detection...
	12. AI returns result...
	13. System updates record...
	14. System displays success

Alternative Courses:

- • A1: Face not detected
- • A2: No blink detected
- • A3: Liveness fails
- • A4: AI Service unavailable

UC-09: Pink Pass Liveness Verification

Identifier	UC-09
Name	Pink Pass Liveness Verification

Priority	High
Actors	AI Service, System
Pre-conditions	Video frames received
Post-conditions	Liveness result returned

Typical Course of Action:

Actor Action	**System Response**
1. System sends video	2. AI receives frames
	3. AI loads Haar Cascade...
	4. AI processes each frame...
	5. If face detected -> CNN...
	6. CNN analyzes EAR...
	7. CNN detects blink...
	8. AI calculates confidence...
	9. AI returns JSON...
10. System logs result	11. System updates user

Alternative Courses:

- • A1: No face detected
- • A2: Multiple faces
- • A3: Blink pattern invalid
- • A4: Gender mismatch

UC-10: Route Deviation Detection

Identifier	UC-10
Name	Route Deviation Detection
Priority	High

Actors	Safety Sentinel, System, Admin
Pre-conditions	Ride active; GPS enabled
Post-conditions	Alert triggered if deviation

Typical Course of Action:

Actor Action	**System Response**
1. Ride begins	2. System retrieves route polyline
	3. Monitoring loop starts...
	4. GPS coords received...
	5. Calc point-to-line dist...
	6. If >500m start timer...
	7. If >30s mark critical...
	8. Trigger Alert (UC-11)

Alternative Courses:

- • A1: Auto-reset
- • A2: GPS lost
- • A3: Manual stop

UC-11: Trigger Safety Alert

Identifier	UC-11
Name	Trigger Safety Alert
Priority	High
Actors	System, Admin, Contacts
Pre-conditions	Anomaly detected
Post-conditions	Notifications sent

Typical Course of Action:

Actor Action	**System Response**
1. Anomaly detected	2. System creates Alert record
	3. System emits Socket.io event
	4. Admin Dash receives event...
	5. Retrieve contacts...
	6. Send Twilio SMS...
	7. Log delivery...
	8. Admin views feed

Alternative Courses:

- • A1: API fail
- • A2: No contacts
- • A3: False alarm cancel

UC-12: Driver Matching Algorithm

Identifier	UC-12
Name	Driver Matching Algorithm
Priority	High
Actors	System, AI Service
Pre-conditions	Ride request created
Post-conditions	Drivers ranked

Typical Course of Action:

Actor Action	**System Response**
1. Passenger confirms	2. System queries drivers
	3. Calc scores (Dist/Rating/Fairness)

	4. Compute Weighted Sum
	5. Sort desc
	6. Return ranked list
	7. Send to top driver

Alternative Courses:

- • A1: No drivers (expand radius)
- • A2: Top driver rejects
- • A3: Pink Pass filter

UC-13: Price Prediction

Identifier	UC-13
Name	Price Prediction
Priority	High
Actors	System, AI Service
Pre-conditions	Route known
Post-conditions	Price displayed

Typical Course of Action:

Actor Action	**System Response**
1. Dest selected	2. Calc dist/duration
	3. Extract features...
	4. Query demand...
	5. Get traffic...
	6. Send to AI...
	7. AI loads model...
	8. Predict price...

	9. Return breakdown...
	10. Display price

Alternative Courses:

- • A1: AI unavailable (fallback)
- • A2: Bounds check

UC-16: Admin Monitor Dashboard

Identifier	UC-16
Name	Admin Monitor Dashboard
Priority	Medium
Actors	Administrator, System
Pre-conditions	Admin logged in
Post-conditions	Status displayed

Typical Course of Action:

Actor Action	**System Response**
1. Navigates to Dash	2. System displays stats
	3. Socket.io connection...
	4. Render Safety Panel...
5. Click Heatmap	6. Request AI heatmap...
	7. AI retrieves locations...
	8. K-Means clustering...
	9. Return clusters...
	10. Overlay on map
11. Click Alert	12. Display details...
	13. Admin action...

Alternative Courses:

- • A1: No alerts
- • A2: Sentiment Analysis

3.3 Requirements Analysis and Modeling

This section includes the analysis models: entity-relationship diagram, abstract class diagram, sequence diagram. As per the project plan, these detailed visual models are documented in the Phase 2 specification. The functional requirements listed above serve as the textual detailed analysis required for this phase.

4. Nonfunctional Requirements

4.1 Performance Requirements

Response Latency: The system is engineered to provide near-real-time responses. Key API endpoints (e.g., login, ride status) must respond within 500ms for 95% of requests under normal load.

Pink Pass Processing: The entire biometric verification process, from video upload to liveness confirmation, must not exceed 15 seconds to minimize user friction.

Driver Matching: The matching algorithm must parse available drivers and return a ranked list within 3 seconds.

Concurrency: The backend infrastructure is scaled to support at least 1000 concurrent active users and handle 100 simultaneous ride sessions without degradation.

4.2 Safety Requirements

Continuous Monitoring: The Safety Sentinel service must maintain an active monitoring loop for every ongoing ride, checking GPS coordinates against the planned route at 5-second intervals.

Alert Delivery: In the event of a detected anomaly (deviation >500m), the system must successfully dispatch an SMS alert to emergency contacts and a Socket.io event to the admin dashboard within 30 seconds.

Data Integrity: All safety-critical data, including route logs and alert timestamps, must be immutable and stored for post-incident auditing.

4.3 Security Requirements

Authentication: All access to the system APIs requires a valid JSON Web Token (JWT), which expires after 24 hours. Passwords must be hashed using bcrypt with a minimum work factor of 10.

Encryption: All sensitive Personally Identifiable Information (PII), specifically CNIC numbers and biometric templates, must be encrypted at rest using AES-256 standard.

Transport Security: All data in transit between the mobile apps and the server must be secured using HTTPS/TLS 1.3 encryption to prevent man-in-the-middle attacks.

4.4 Additional Software Quality Attributes

Usability: The mobile applications shall adhere to Material Design (Android) and Human Interface Guidelines (iOS) to ensure intuitive navigation.

Reliability: The system targets a 99% uptime availability during operational hours (6 AM to 12 AM). Critical safety services are designed with redundancy to achieve 99.9% reliability.

Maintainability: The codebase shall follow standard style guides (ESLint for JS, PEP 8 for Python) to facilitate future updates. API documentation shall be maintained via Swagger.

5. Other Requirements

Legal Compliance: The system shall incorporate features to maximize alignment with local transport regulations, including digital logs of all trips.

Localization: While the initial interface is English, the architecture shall support internationalization (i18n) to allow for Urdu language support in future updates.

Hardware Compatibility: The application must remain functional on mid-range smartphones common in the target market, optimizing resource usage to prevent overheating or excessive battery drain.

6. Revised Project Plan

The SAFORA project adheres to an Agile Scrum methodology, broken down into 2-week sprints. The current status puts the team in Phase 2 (Backend Core Development). The roadmap for the upcoming weeks includes:

Weeks 6-8: Completion of AI Microservice (Pink Pass & Pricing Models).

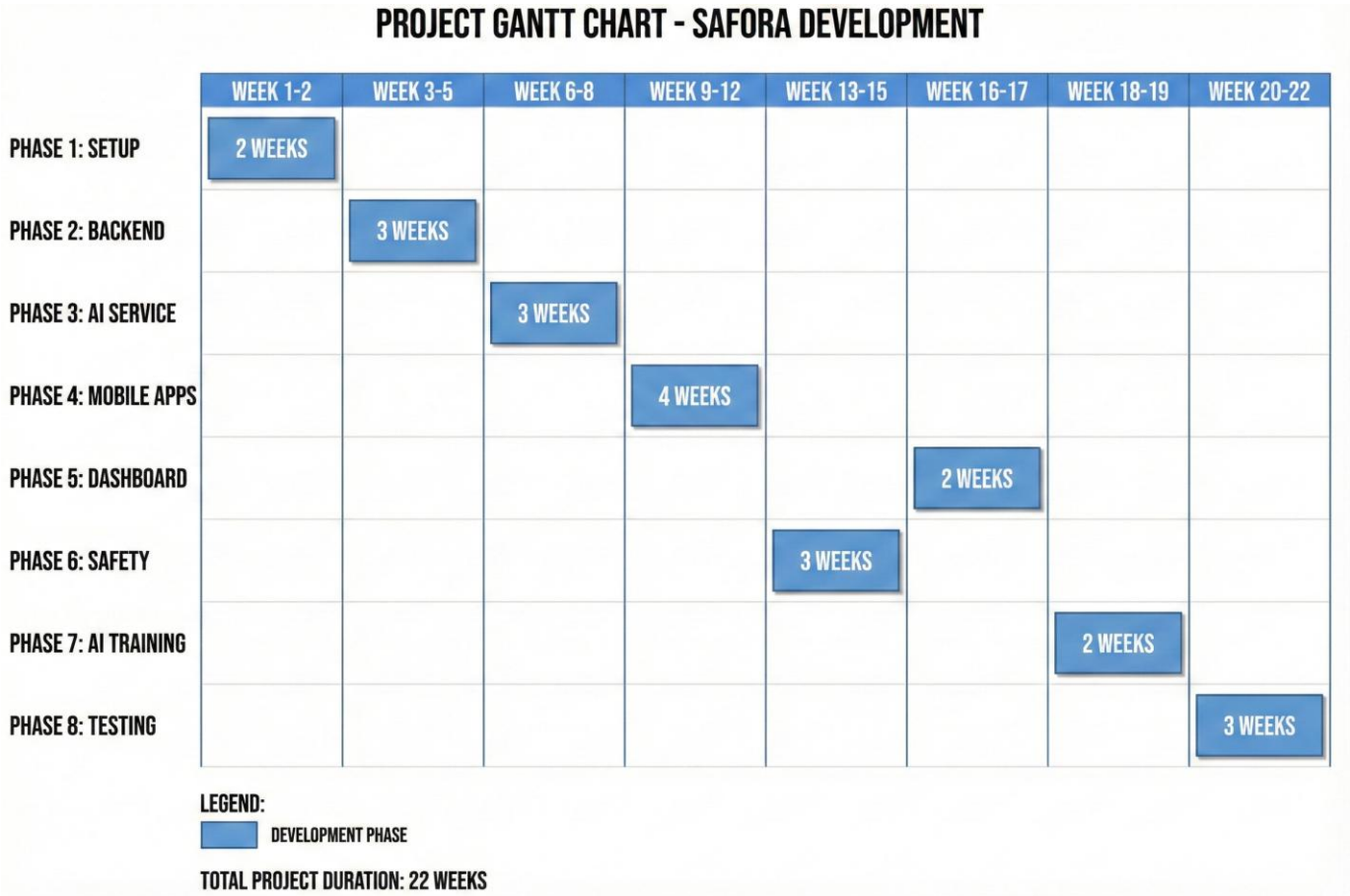
Weeks 9-12: Development of cross-platform Mobile Applications.

Weeks 13-15: Integration of Safety Features (Safety Sentinel).

Weeks 16-17: Admin Dashboard implementation.

Weeks 18-22: Extensive Testing, QA, and deployment preparation.

This iterative approach allows for continuous feedback and refinement of features based on testing results.



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Appendix A: Glossary

The following terms are defined to ensure a consistent understanding of this SRS document:

Term/Acronym	Definition
AI (Artificial Intelligence)	Simulation of human intelligence processes by machines, used here for decision making in driver matching and pricing.
API (Application Programming Interface)	A set of definitions and protocols for building and integrating application software.
CNN (Convolutional Neural Network)	A class of deep neural networks, most commonly applied to analyzing visual imagery, used for Pink Pass verification.
CNIC	Computerized National Identity Card, utilized for robust identity verification in Pakistan.
EAR (Eye Aspect Ratio)	A scalar quantity computed from eye landmarks to detect blinking in liveness detection.
GPS (Global Positioning System)	Satellite-based navigation system used for tracking vehicle location.
JWT (JSON Web Token)	An open standard (RFC 7519) that defines a compact and self-contained way for securely transmitting information between parties as a JSON object.
Pink Pass	A proprietary feature of SAFORA offering biometrically verified, women-only rides.
Safety Sentinel	The automated background service responsible for monitoring active rides for route deviations.
Socket.io	A library that enables real-time, bidirectional and event-based communication between the browser and the server.

Appendix B: IV & V Report

(Independent Verification & Validation)

IV & V Resource

Name	Signature
	_____

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
4					
5					

List of non-trivial defects